



Customer Shopping Behavior Analysis

Transaction data from **3,900 purchases** across multiple product categories — uncovering spending patterns, preferences, and revenue drivers.

PYTHON · POSTGRESQL · POWER BI

Dataset at a Glance

3,900

Purchase Records

18

Features

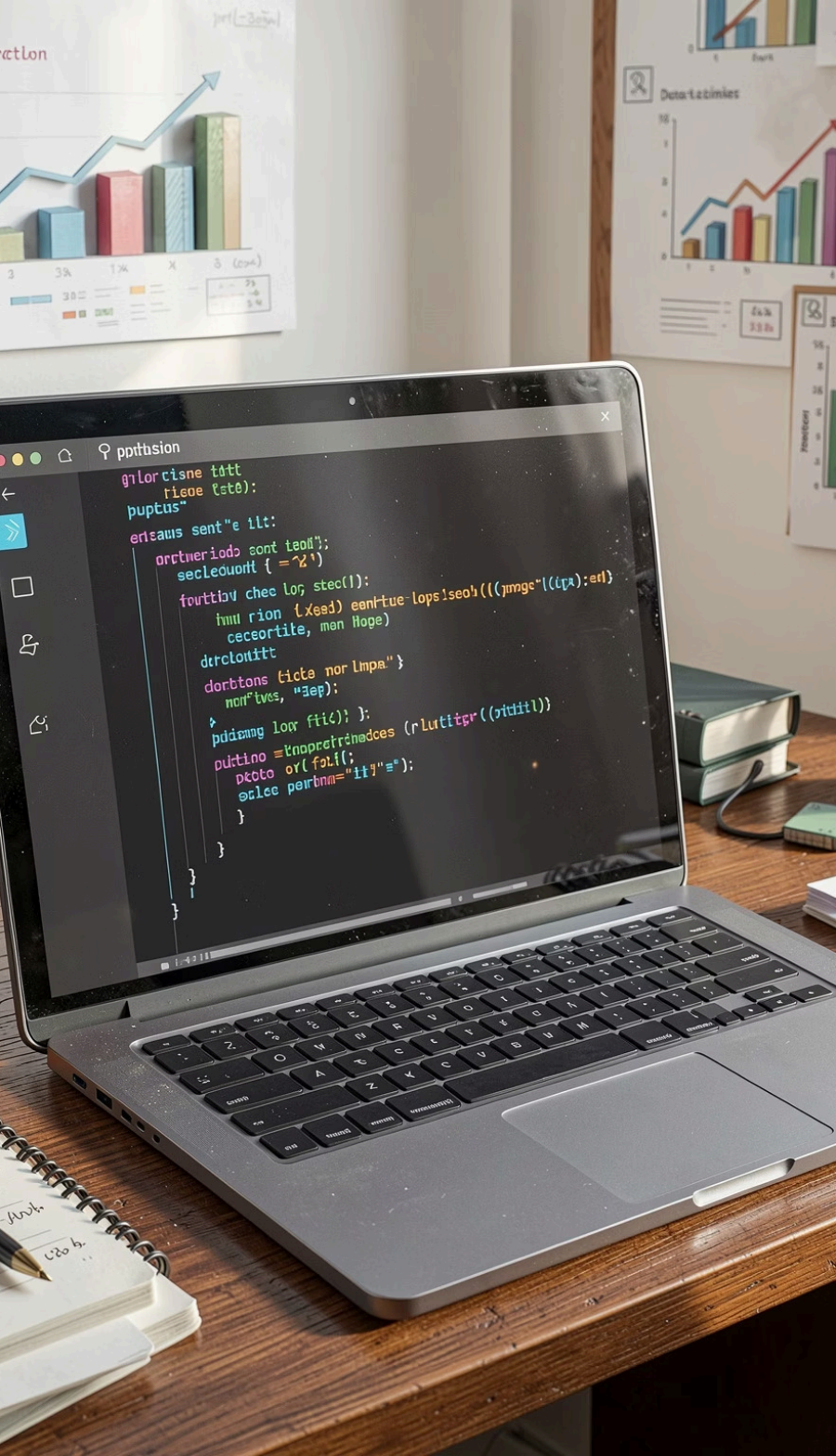
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Product Categories

\$59.76

Avg Purchase

Demographics · Product info · Transaction details · Engagement metrics



Data Preparation in Python

01

Load & Explore

Pandas: structure, types, stats

02

Clean & Standardize

Missing values, snake_case

03

Feature Engineering

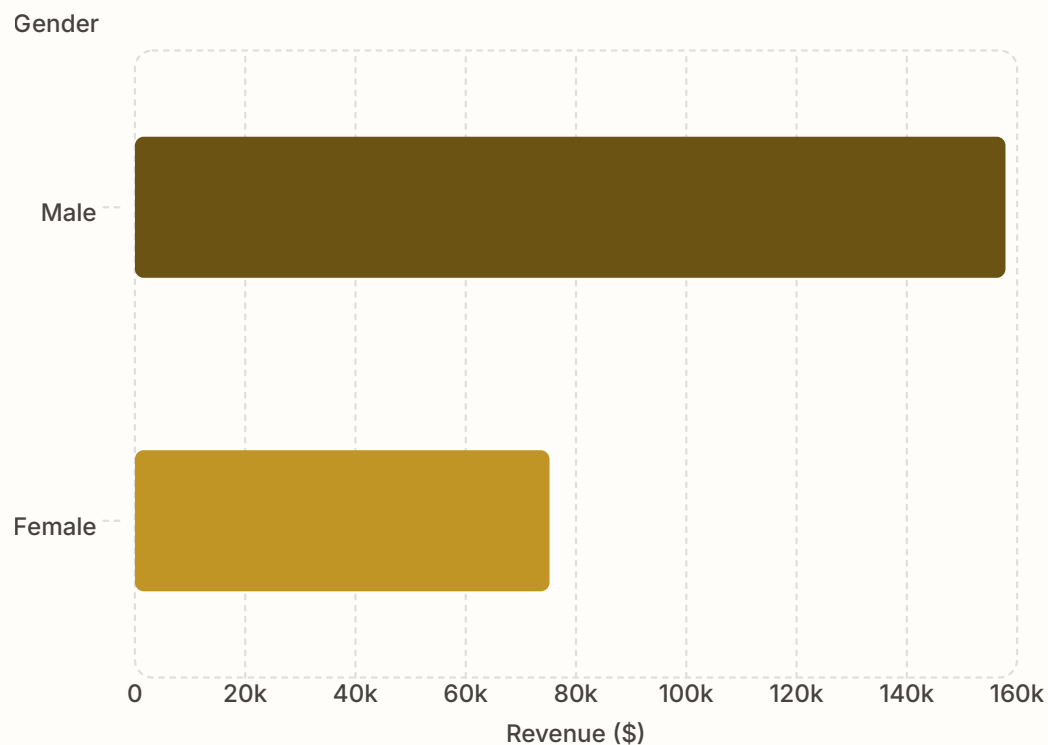
Age groups, purchase frequency

04

Export to PostgreSQL

Ready for SQL & BI

Revenue & Spending Insights



Key Findings

- Male revenue **2× higher** than female
- 839 discount users spent above average
- Express shipping: **\$60.48** vs Standard **\$58.46**

Top Products & Discounts

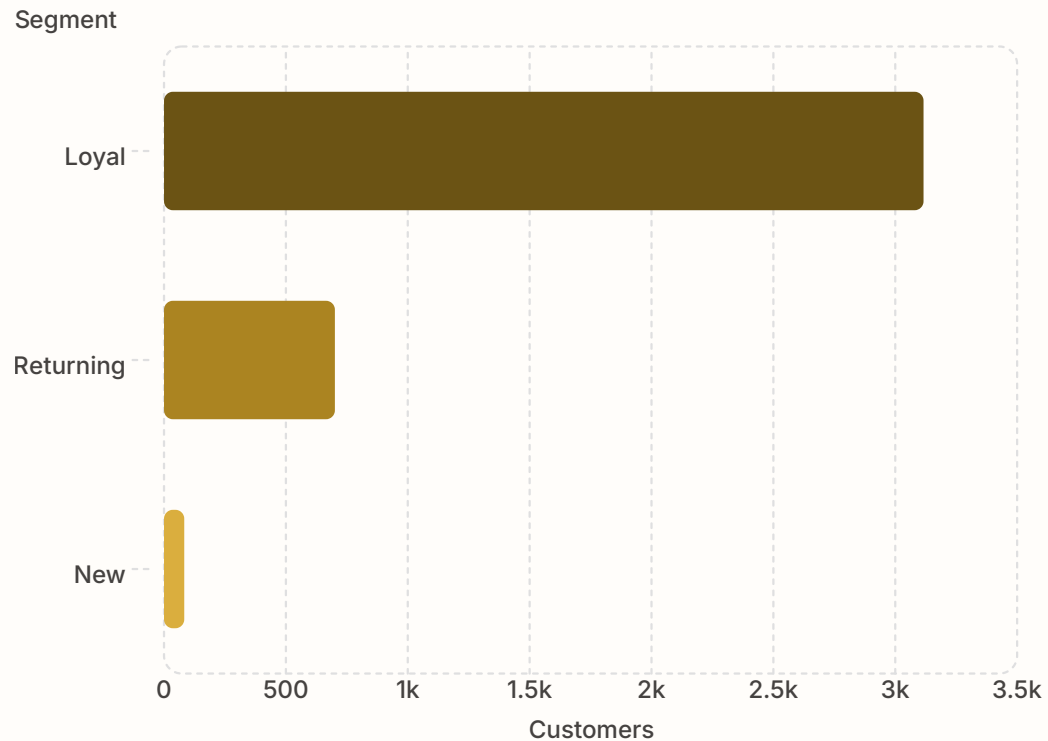
Top-Rated Products

Item	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79

Highest Discount Usage

Item	Discount Rate
Hat	50%
Sneakers	49%
Coat	49%
Sweater	48%
Pants	47%

Customer Segmentation

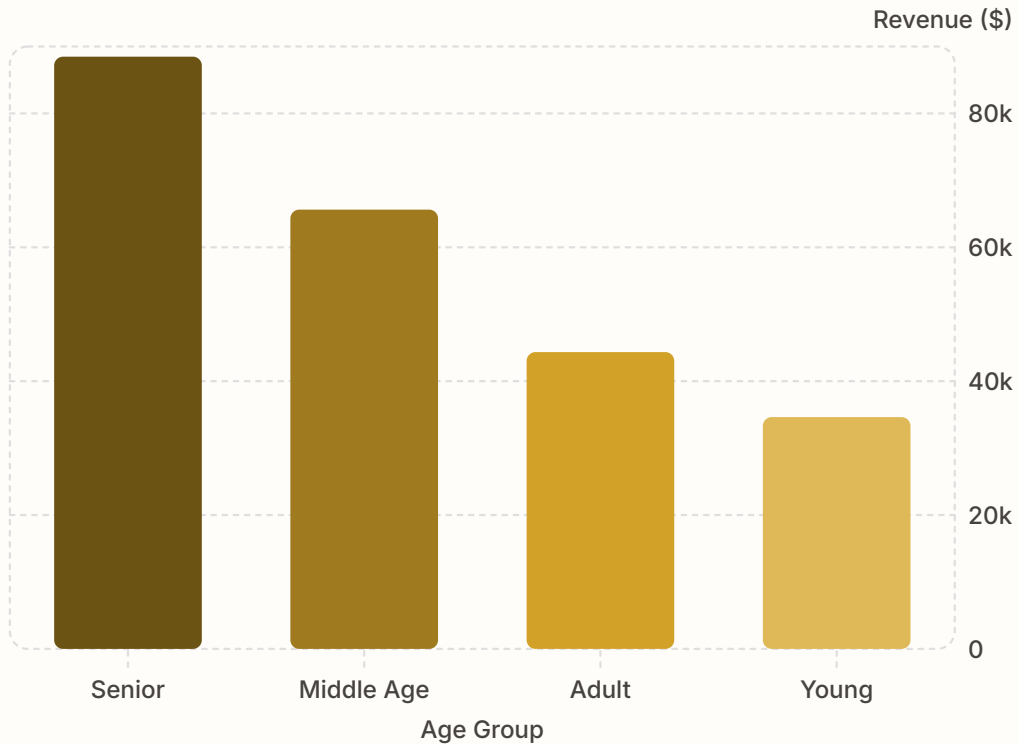


Segment Breakdown

- **Loyal:** 3,116 customers — dominant segment
- **Returning:** 701 customers
- **New:** 83 customers — growth opportunity

📋 Repeat buyers: 2,518 non-subscribers vs 958 subscribers

Revenue by Age Group

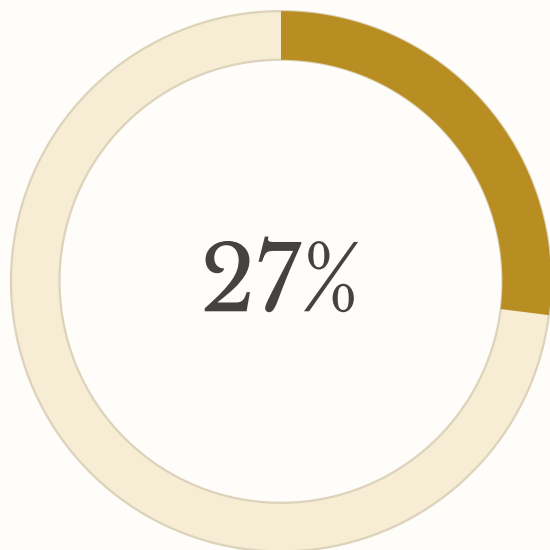


High-Value Segments

- **Seniors** lead revenue at \$88,480
- Middle Age: \$65,629
- Young customers: lowest at \$34,630

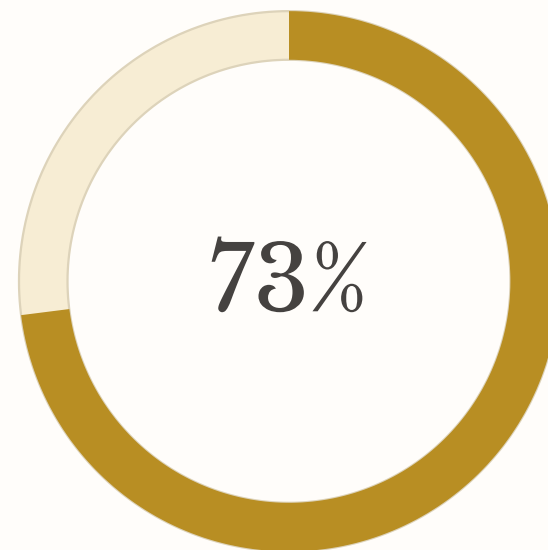
Target personalized campaigns at top age groups.

Subscription Analysis



Subscribers

1,053 customers · \$62,645 revenue



Non-Subscribers

2,847 customers · \$170,436 revenue

i Non-subscribers generate **2.7× more total revenue** — major growth opportunity through subscription adoption.

Power BI Dashboard

Interactive dashboard with KPIs, filters, and visualizations for data-driven decisions.



Business Recommendations



Boost Subscriptions

Exclusive benefits for non-subscribers



Focus Top Categories

Invest in Footwear & Clothing



Target High-Value

Campaigns for Seniors & Middle Age



Retention & Loyalty

Loyalty programs for repeat buyers



Optimize Discounts

Targeted discounts, protect margins



Leverage Analytics

Insights for marketing decisions